

120 Days of Quantitative Finance

Month 3: Machine Learning for Finance

Week 11: Portfolio Optimization and Risk Management using ML

Day 72: Limitations of Traditional Portfolio Management and Motivation
for Machine Learning

1 Introduction

Traditional portfolio management techniques are largely based on the framework developed by Harry Markowitz, known as Modern Portfolio Theory (MPT).

While MPT provides a mathematically elegant approach to diversification and risk management, its practical implementation faces several challenges in real financial markets.

These limitations have motivated the use of machine learning techniques in modern portfolio management.

2 Dependence on Expected Returns

One of the key inputs in portfolio optimization is the vector of expected asset returns:

$$\mu = (\mu_1, \mu_2, \dots, \mu_n)$$

However, estimating expected returns is extremely difficult in financial markets.

Small estimation errors in expected returns can lead to large changes in optimal portfolio weights.

This phenomenon is known as **estimation error amplification**.

3 Unstable Covariance Estimates

Portfolio risk depends on the covariance matrix of asset returns:

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \dots & \sigma_{nn} \end{bmatrix}$$

In practice, covariance matrices estimated from historical data are often unstable and sensitive to the chosen time period.

During market crises, correlations between assets tend to increase, reducing the effectiveness of diversification.

4 Assumption of Stationarity

Traditional models assume that statistical properties of asset returns remain constant over time.

In reality, financial markets exhibit:

- Changing volatility
- Structural breaks
- Regime shifts

This non-stationarity makes historical estimates less reliable for predicting future behavior.

5 Linear Model Assumptions

Classical portfolio models rely on linear relationships between variables.

However, financial markets often exhibit nonlinear interactions between factors such as:

- macroeconomic variables
- market sentiment
- liquidity conditions

Linear models may fail to capture these complex relationships.

6 High-Dimensional Data Challenges

Modern financial datasets include large numbers of assets and features.

When the number of assets approaches the number of observations, covariance estimation becomes unstable.

This is sometimes referred to as the **curse of dimensionality**.

7 Motivation for Machine Learning

Machine learning methods can help address many of these limitations.

Key advantages include:

- Ability to model nonlinear relationships
- Improved handling of high-dimensional datasets
- More flexible prediction models
- Data-driven feature discovery

Machine learning techniques are increasingly used for:

- return prediction
- risk forecasting
- portfolio optimization
- dynamic asset allocation

8 Examples of ML Techniques in Portfolio Management

Several machine learning methods are commonly used in quantitative finance:

- Regularized regression models
- Tree-based ensemble models
- Neural networks

- Reinforcement learning

These approaches allow portfolio managers to incorporate a wider range of data sources and model more complex patterns.

9 Conclusion

While traditional portfolio theory remains an important foundation in finance, its assumptions often fail in real-world markets.

Machine learning offers powerful tools to overcome these limitations by enabling more flexible, data-driven models.

In the following sections, we will explore how machine learning techniques can be applied to portfolio optimization and risk management.